15.4 Comparing Probabilities

Lesson Introduction

Benchmark	MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.	Learning Targets	• I can interpret the likelihood of a chance event occurring using probability. • I can compare probabilities of chance events.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How can you express the likelihood of a chance event as a probability? • How can you compare probabilities expressed in different forms?
Lesson Narrative	In this lesson, students will practice determining probability of an event, describing its likelihood, and comparing it to the probabilities of other events. Students will complete a Card Sort comparing and describing probabilities of chance events before practicing on their own. To conclude the lesson, students compare two student responses when interpreting and comparing probabilities of two events.

Component Summary				
	15.4.1 Warm-Up (~5 min)	Career Video – Statistician (SE pg. 417)	15.4.3 Guided Practice (10 min)	Practice interpreting and comparing probabilities (SE pg. 418)
	15.4.2 Activity (15 min)	Probability Card Sort (SE pg. 417)	15.4.4 Your Turn! (5–10 min)	Practice ordering probabilities in different forms (SE pg. 419)
	15.4.5 Wrap-Up (~5 min)	Compare two student responses to determine who is correct (SE pg. 420)		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Sample space • Chance • Probability • Likelihood	• Career Profile Video • Cards for 15.4.2 Activity	• 15.4.1 Warm-Up contains a video to accompany the question prompts.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may get confused with probability and likelihood. The scaffolding in this lesson first has students order events by probability and then describe the likelihood.
Students may need support with converting between fractions, percentages, and decimals.</td><td>Consider ways to make connections among describing the likelihood of events based on their probability throughout the lesson.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
In the 15.4.2 Activity, after students order their first set of cards with their partner, students can compare/discuss with another group to see how they ordered their cards before continuing with the next set of cards.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Students may have experience with seeing statisticians in the real world, like watching sports and hearing commentators share statistics about players or teams.
Statistics have many applications in a variety of career fields that are not typically thought of as STEM. Use 15.4.1 Warm-Up as an opportunity to highlight this and the importance of understanding statistics in the real world to avoid manipulation.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider a kinesthetic approach to 15.4.3 question 2. Read the problem aloud and have students move to one side of the room or the other depending on which statement they think is more likely. Students who think the statements are equally likely can stand in the middle of the room.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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15.4.1 Warm-Up: Statisticians Careers

Component Overview: *Play the Career Profile video on the Statistician for the entire class. After students watch the video (2:06 minutes), prompt them to reflect on the profession by answering the questions independently.*

15.4.1 Warm-Up: Statisticians Careers

Statisticians turn data into understandable and useful information. The government, health care industry, sports broadcasting, and many businesses have statisticians who analyze data to predict trends. Statisticians help businesses and industries with decision-making based on analysis.

1. What are some real-world examples where you have seen statistics used?
   Answers will vary.

2. Consider an industry you are interested in working in someday.

   a. What industry are you considering?
      Answers will vary.

   b. What is at least one way you could see trends in data and analysis of statistics used in this industry?
      Answers will vary.

MA.K12.MTR.7.1: Mathematics in the Real World
Students may have experience with seeing statisticians in the real world, like watching sports and hearing commentators share statistics about players or teams. Statistics has many applications in a variety of career fields that are not typically thought of as STEM. Use this Warm-Up as an opportunity to highlight this and the importance of understanding statistics in the real world to avoid manipulation.

15.4.2 Activity: Likelihood of Chance Events

Component Overview: *Students will work with a partner to order the events on the Card Sort from least likely to most likely. After sorting the cards, students work with their partners to answer questions where they describe the likelihood of events or express them using probabilities.*

15.4.2 Activity: Likelihood of Chance Events

Work with your partner or group to order the events on the cards from least likely to most likely.

After ordering the events, work with your partner or group to complete the following using the events from the card sort.

1. Explain which event(s), if any, can be described as "impossible."
   Answers will vary. These will be the events that have a 0% chance of happening.
   - *Drawing out a ball with the number 3 from a fishbowl containing balls with even numbers written on them*

2. Explain which event(s), if any, can be described as "certain."
   Answers will vary. These will be the events that have a 100% chance of happening.
   - *Choosing a block with 4 sides of the same length from a pile of square blocks*

3. Explain which event(s), if any, can be described as "equally likely as not."
   Answers will vary. These will be any event that has a 50% chance of happening or not.
   - *Pulling a red card from a deck with half the cards red*

4. Express the likelihood of a randomly chosen word in a novel beginning with the letter T, as a percentage.
   $\frac{4}{25} = 0.16; \quad 0.16 \cdot 100 = 16\%$

5. Express the likelihood of a randomly selected person being left-handed, as a decimal.
   $10\% = \frac{10}{100} = 0.1$

Consider using the two sets of cards separately so that you can monitor for understanding and address any common misconceptions after the first set before giving the next set.

Fast finishers can go through and express the probability of each even number in the Card Sort.

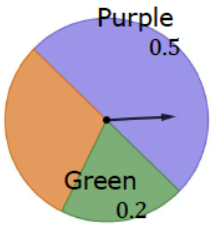
15.4.3 Guided Practice: Interpreting and Comparing Probabilities

Component Overview: *Students practice interpreting and comparing probabilities of chance events.*

15.4.3 Guided Practice: Interpreting and Comparing Probabilities

1. A spinner is shown with the probabilities of landing on each space given as a decimal.

   Use the spinner to complete the table by identifying the missing events, probabilities, and descriptions of likelihood.



Event	Probability	Likelihood
$P(\text{purple})$	50%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{orange})$	30%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{yellow})$	0%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{green})$	20%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{purple or orange})$	80%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{color})$	100 %	o impossible o unlikely o equally likely as not o likely o certain

Consider prompting students to discuss the possible outcomes of spinning the spinner, including the likelihood of landing on each color prior to having them engage with the table on the next page.

Students can create their own spinner and table and determine the probability and likelihood of the events in the spinner.

15.4.3 Guided Practice: Interpreting and Comparing Probabilities (continued)

2. For each scenario, circle the event that is more likely. If the events are equally likely, circle both.

   a. Alex and Elvis play baseball. Based on their statistics, Alex's probability of hitting a home run is 0.27. The probability of Elvis hitting a home run is 14%. Which choice is more likely on their next at bat?

Alex hits a home run.	or	Elvis hits a home run.

   b. A meteorologist states there is a 50% chance for rain tomorrow. Which choice is more likely to happen?

It will rain tomorrow.	or	It will not rain tomorrow.

   c. Shawn is calling the coin toss before his football game. Which choice is more likely to occur?

The coin will land on its edge.	or	The coin will land on either heads or tails.

Consider a kinesthetic approach to this question. Read the question aloud and have students move to one side of the room or the other depending on which statement they think is more likely. If they think they are equal, they can stand in the middle of the room.

15.4.4 Your Turn!

Component Overview: *Students practice comparing and ordering the probabilities of chance events expressed in different forms.*

15.4.4 Your Turn!

1. The probability that Lawrence makes a free throw when playing basketball is 60%. The likelihood that he makes a 3-point shot is 0.345.

   Explain which event is more likely, Lawrence making a free throw or making a 3-point shot.

   The event that is more likely is that Lawrence will make a free throw because the likelihood is 60%. Converting the probability that he will make a 3-pointer (0.345) to a percentage is 34.5% which is less the 60% for the free throw.

2. The probabilities of five events are shown.

   60% 8 out of 10 0.37 20% $\frac{5}{6}$

   List the probabilities in order from least to greatest.

   20%, 0.37, 60%, 8 *out of* 10, $\frac{5}{6}$

How can you compare probabilities in different forms?

15.4.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students compare two student responses to determine who is correct when comparing the likelihood of two different events.*

15.4.5 Wrap-Up: Ticket Out the Door

1. There are 25 prime numbers between 1 and 100. There are 46 prime numbers between 1 and 200. Two events are shown.

Event A	Event B
A computer produces a random number between 1 and 100 that is prime.	A computer produces a random number between 1 and 200 that is prime.

Ava said Event B is more likely to occur because there are 46 possible outcomes in the event compared to only 25 in Event A. Oscar said Event A is more likely because $P(\text{A}) = 0.25$ and $P(\text{B}) = 0.23$.

a. Explain who is correct.
Oscar is correct because the probability for Event A is $P(A) = 0.25$ because $\frac{25}{100} = 0.25$ and the probability for event b is $P(B) = 0.23$ because $\frac{46}{200} = 0.23$.

b. Write a short note to the student who was incorrect explaining the error they made.
Answers will vary. Ava, probability is the ratio of favorable outcomes to total outcomes. It is not enough to simply compare the number of favorable outcomes without comparing to the total possible outcomes.

The responses provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.